

Big-O Proofs with Inductionex. Prove 2^n is $O(n!)$

$$j! = j(j-1)! \geq 2^j$$

$$(j-1)! \geq 2^{j-1}$$

$$2(j-1)! \geq 2^j$$

Choose $c=1$, $k=4$, by big-O definition, we need to show $2^n \leq 1 \cdot n!$ for $n \geq 4$. We will prove by induction on n . For base case, $n=4$, $2^4 \leq 4!$, $16 \leq 24$, so base case holds. For IH, suppose $2^n \leq n!$ for $n=4, 5, 6, \dots, j-1$. For inductive step, $n=j$ and we need to show that $2^j \leq j!$ or $j! \geq 2^j$. By IH, $(j-1)! \geq 2^{j-1}$, $(j-1)! \geq 2^{j-1}$. Then $2^j \leq 2(j-1)! \leq j(j-1)!$ (bc $j \geq 4 > 2$) = $j!$, so $2^j \leq j!$ and claim is proven, 2^n is $O(n!)$

Algorithms

Accessing basic data structures:

• arrays

- element access $O(1)$ - changing length/adding/deleting $O(n)$

• linked list

- adding/removing/reading/writing from head or constant point $O(1)$ if linked list w/ length n , accessing 5th element is $O(5) = O(1)$, constant cuz not dependent on n - add/remove/read/write from tail or point reliant on n $O(n)$ eg middle of list, $n/2$ element, $O(\frac{n}{2}) = O(n)$ Pseudocode

ex. Iterative algorithm closestpair

ask if depends on length of list $O(1)$

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01 closestpair( $p_1, \dots, p_n$ ) : array of 2D points)
02   best1 =  $p_1$ 
03   best2 =  $p_2$ 
04   bestdist = dist( $p_1, p_2$ )
05   for i = 1 to n → all points
06     for j = 1 to n → all points } all pairs of points
07       newdist = dist( $p_i, p_j$ )
08       if ( $i \neq j$  and newdist < bestdist)
09         best1 =  $p_i$ 
10         best2 =  $p_j$ 
11         bestdist = newdist
12   return (best1, best2)

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$O(n^2)$ $O(n) \times O(n)$ $O(1)$

Common exam questions:

1) What is the algorithm doing?

returns two points in the array closest to each other

2) What is the big-O running time?

02-04: $O(1)$

$$05-11: O(n^2)$$

$$12: O(1)$$

$$\text{total: } O(1+n^2+1) = O(n^2), \Theta(n^2)$$

Discussion Problems - Induction Study Problems

Prove the following claim by strong induction:

$$2^{k-1}(k-1)! < (2(k-1))!$$

$$2k! = 2k(k-1)!$$

Claim: For all positive integers $n \geq 2$, $2^n n! < (2n)!$

$$2^{k-1} < 2$$

Proof: Prove by induction on n

Base case: $n=2$, $2^2(2!) < (2(2))!$ $4 \cdot 2 < 4!$ $8 < 24$, base case holds

IH: $2^n n! < (2n)!$ for $n=2, 3, 4, \dots, k$

IS: Goal to show $n=k+1$, $2^{k+1}(k+1)! < (2(k+1))!$. By IH, we know $2^k k! < (2k)!$

therefore $(2(k+1))! = (2k+2)! = (2k+2)(2k+1)(2k)! = (2k+1)[2(k+1)](2k)!$

$> (2k+1)[2(k+1)]2^k k! = (2k+1)2^{k+1}(k+1)!$. Since $k \geq 2$, $(2k+1) > 1$, so

$(2k+1)2^{k+1}(k+1)! > 2^{k+1}(k+1)!$, therefore $(2k+2)! > (2k+1)2^{k+1}(k+1)! > 2^{k+1}(k+1)!$,

$2^{k+1}(k+1)! < (2k+2)!$, $\square$